One Billion Hungry

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Can We Feed the World?

GORDON CONWAY
with Katy Wilson

Foreword by Rajiv Shah

Published with the support of
Agriculture for Impact, Imperial College, London

Comstock Publishing Associates, a division of
Cornell University Press | Ithaca and London

This publication is based on research funded in part
by the Bill & Melinda Gates Foundation.
The findings and conclusions contained within
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For information, address Cornell University Press,
Sage House, 512 East State Street, Ithaca, New York 14850.

First published 2012 by Cornell University Press
First printing, Cornell Paperbacks, 2012

Printed in the United States of America and the United Kingdom

Library of Congress Cataloging-in-Publication Data
Conway, Gordon.
One billion hungry : can we feed the world? / Gordon Conway with
Katy Wilson ; foreword by Rajiv Shah.
p. cm.
"Published with the support of Agriculture for Impact, Imperial College, London."
Includes bibliographical references and index.
ISBN 978-0-8014-5133-1 (cloth : alk. paper)
ISBN 978-0-8014-7802-4 (pbk. : alk. paper)
1. Agricultural innovations—Developing countries. 2. Green Revolution—
Developing countries. 3. Food supply—Developing countries.
4. Food security—Developing countries. I. Wilson, Katy.
II. Conway, Gordon. Doubly green revolution. III. Title.

S494.5.I5C663 2012
338.1'6091724—dc23

2012014403

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Cloth printing 10 9 8 7 6 5 4 3 2 1
Paperback printing 10 9 8 7 6 5 4 3 2 1

To the memory of
Cyril and Thelma Conway

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Foreword

by Rajiv Shah

Administrator of the United States Agency for International Development

By the late 1990s, global food security had mostly fallen off the world's agenda. The success of the Green Revolution had helped hundreds of millions of people in Latin America and Asia avoid a life of extreme hunger and poverty. Governments—developed and developing alike—assumed this success would spread and cut their investments in agriculture, allowing them to turn their attention elsewhere.

But while many had lost sight of the importance of agricultural development, Gordon Conway stayed focused. In his book, *A Doubly Green Revolution*, published in 1997, Gordon issued a pressing call for the development community to recommit to the goals of fighting hunger and malnutrition around the world.

Gordon argued that the lasting elimination of hunger required us to do more than transform the production of food—the hallmark achievement of the Green Revolution. We also had to help smallholder farmers build resilience to natural disasters and climate change, use advances in science and technology to boost yields, and partner with the private sector to get those crops to market. Thanks to Gordon's leadership, the world is once again delivering a global commitment to strengthening food security.

In 2009, President Obama established a global food security initiative called Feed the Future. Spearheaded by the U.S. Agency for International Development, Feed the Future helps countries develop productive agriculture sectors so they can feed themselves and transform their economies over the long term. In partnership with local and international private companies, the global initiative supports country-led plans to focus investment in regions most likely to flourish, and in crop and livestock systems with the greatest chance of fighting poverty and malnutrition.

These efforts are more important now than ever before. In the Horn of Africa, the worst drought in 60 years put more than 13.3 million people in the path of hunger and disease during 2011. In Somalia, where decades of civil war and disorder have contributed to the complete breakdown of governance, drought led to a famine that threatened the security and economic growth of the entire region.

In coordination with humanitarian assistance, long-term approaches to strengthen resilience—for example, safety nets, frontline health programs, and vaccination campaigns—helped mitigate the worst of the drought impacts across the Horn. But

the reality is that we must do more as a global community to prevent these crises from occurring in the first place.

We have to harness the power of science and technology to transform agriculture around the world. Our efforts must begin by strengthening investments in breakthrough technologies, such as climate-resilient seeds, and breakthrough services, such as climate-based agricultural insurance.

In today's difficult growing conditions, smallholder farmers from Kenya to India are improving their yields with new varieties of drought- and disease-resistant seeds and other technologies. To continue the development of crop and livestock advances that boost farm income and improve nutrition, we must support the full range of research techniques, including both conventional and genetically engineered approaches, and look beyond the issues centered on production alone.

An invaluable voice in the fight against hunger, Gordon's new book, *A Billion Hungry: Can We Feed the World?*, calls on the global community to do more—guiding our way forward while building on the messages of *The Doubly Green Revolution*. As global population grows toward 10 billion people, the challenges ahead remain daunting. Nonetheless, strong community leadership, renewed international commitment, and new opportunities, from mobile technology to drought-resistant seeds, are dramatically expanding the possibilities for success. By delivering meaningful results more effectively and efficiently than ever before, we can share Gordon's optimism that we will finally end hunger.

Acknowledgments

In some respects this is a second edition of my book *The Doubly Green Revolution: Food for All in the 21st Century,* first published in 1997 by Penguin Books in the United Kingdom and then a year later by Cornell University Press (and in a Portuguese translation, *Produção de Alimentos no Século XXI: Biotecnologia e Meio Ambiente,* Estação Liberdade, São Paulo, Brazil in 2003).

As in the previous edition, the book draws on a wide variety of sources and is indebted to the work of those with whom I have been associated over the past fifty years. Many are named in the text. They include postgraduate students and faculty at the Imperial College of Science and Technology, the University of Sussex, the Universities of Chiang Mai and Khon Kaen in Thailand, Padjajaran University, Indonesia, and the University of the Philippines, Los Baños, Philippines; colleagues at institutes where I have worked or been a director—the International Institute for Environment and Development (IIED), the Institute of Development Studies (IDS), the International Food Policy Research Institute (IFPRI), the Ford and Rockefeller Foundations, the UK Department for International Development (DFID); and workers in numerous nongovernmental organizations including the Aga Khan Rural Support Programme (AKRSP) in Pakistan and India, Action Aid and Mysore Relief and Development Agency (MYRADA) in India, Winrock International in Nepal, Concern Worldwide, the Ethiopian Red Cross, the Mo Ibrahim Foundation, the Alliance for a Green Revolution in Africa (AGRA), and the African Agricultural Technology Foundation (AATF); and the many scientists at the International Agricultural Research Centers (IARCs) of the Consultative Group on International Agricultural Research (CGIAR).

I have drawn freely on their writings and am grateful for their advice, experience, and friendship. Perhaps the greatest influence on my thinking has been the group of scientists at the University of Chiang Mai, Thailand, notably Benjavan Rerkasem, Kanok Rerkasem, Phrek Gypmontasiri, Rapeepan Jaisaard, Methi Ekasingh, Manu Seetisarn, Nakorn Na Lampang, and Ian Craig.

I am also particularly indebted to Tim Wheeler at DFID, Prabhu Pingali at the Bill & Melinda Gates Foundation, Rattan Lal at Ohio State University, Lawrence Haddad at the Institute for Development Studies, Peter Hazell at SOAS and visiting professor at Imperial College London, Margaret Catley-Carlson at CGIAR, Gary Toenniessen at the Rockefeller Foundation, Christopher Delgado at the World

Bank, and K. L. Heong at IRRI, who have read and commented on parts of the book.

The conception of the *Doubly Green Revolution* was an outcome of the deliberations of a small panel, which I chaired, commissioned to develop a vision statement for the CGIAR—the organization that supports the international research centers, which, over the past fifty years, have provided the research base that spearheaded not only the original Green Revolution (detailed in Chapter 3) but attempts to replicate its successes.[1] The vision was presented and adopted at a meeting of ministers of overseas development from the developed countries and of agriculture and natural resources from the developing countries, held in Lucerne in February 1995. My colleagues on the panel were Uma Lele, of the University of Florida, Martin Piñeiro (formerly director of the Inter-American Institute for Cooperation on Agriculture [IICA]), Jim Peacock, Chief of the Division of Plant Industry of Australia's Commonwealth Scientific and Industrial Research Organization (CSIRO), Selçuk Özgediz of the World Bank, Johan Holmberg of the Swedish Agency for Research Cooperation with Developing Countries (SAREC), Henri Carsalade of the French Centre for International Cooperation in Development-Oriented Agricultural Research (CIRAD), Michel Griffon of CIRAD, and Peter Hazell of IFPRI. I am grateful to all of them, to Paul Egger of the Swiss Development Cooperation and Robert Herdt of the Rockefeller Foundation, both on the CGIAR's Oversight Committee, and to Ismail Serageldin, then chair of CGIAR, for their collegial support and encouragement. Many of the ideas in this book are theirs.

In 1998, not long after *Doubly Green Revolution* was published in the United States, I became the twelfth president of the Rockefeller Foundation, centered in New York. The Foundation has a distinguished history in funding agricultural research and development, going back to the origins of the Green Revolution in the early 1940s (which I describe in Chapter 3). During my tenure as president we continued to fund pioneering programs in agriculture, under the direction of first-class international staff led by Bob Herdt, Gary Toenniessen and Peter Matlon. The team included Joe DeVries, Akin Adesina, Ruben Puentes, John O'Toole, Bharati Patel, and John Lynam.

I returned to the United Kingdom at the end of 2004 and became the chief scientific adviser to the UK Department for International Development (DFID). Again this was, and still is, an organization supplying major funding to agricultural research. I benefited from the experience of many of the DFID staff, in particular Jonathan Wadsworth, Alan Tollervey, and Steve Hillier, and the chief economist Tony Venables.

I have also greatly benefited from Sir John Beddington, the UK government chief scientist; a former holder of the post, Lord Robert May; and another departmental

chief scientist, Sir Roy Anderson. I have known all three of them for many years and been stimulated by their ideas.

While at DFID I helped to set up the UK Collaborative on Development Sciences (UKCDS) under the very able direction of Andrée Carter. She was also heavily involved in the production of a book entitled *Science and Innovation for Development,* authored by Jeff Waage, of the London International Development Centre, Sara Delaney, and myself, published by the UKCDS and funded by DFID (free to download from the UKCDS website). I have drawn heavily from some of its contents. Once it was complete, Sara Delaney spent several months working with me on the early drafts of this book.

I stepped down from the chief scientist post in 2010 and returned to Imperial College as Professor of International Development running an advocacy grant from the Bill & Melinda Gates Foundation. Under the grant, we have set up a Montpellier Panel consisting of distinguished European and African agricultural experts: Tom Arnold, Joachim von Braun, Henri Carsalade, Louise Fresco, Peter Hazell, Namanga Ngongi, David Radcliffe, Lindiwe Majele Sibanda and Ramadjita Tabo. I have gained much from their wisdom, and also from the leaders of a sister Gates Foundation grant, the Global Agricultural Development Initiative of the Chicago Council for Global Affairs, Marshall Bouton, Catherine Bertini, and Dan Glickman; and from the staff of the Gates Foundation, Brantley Browning, Mark Suzman, Joe Cerell, and Laurie Lee from the advocacy program and Rajiv Shah, Sam Dryden, Prabhu Pingali, Rob Horsch, Roy Steiner, and Lutz Goedde from the agriculture program.

I benefited in 2011 from a stay at the Rockefeller Foundation's villa at Bellagio on Lake Como in Italy, accompanying my wife, Susan, who was working on her book on the Shan. It provided much-needed time to dedicate to this book at a critical juncture. I also owe much to Susan for her encouragement and sound advice.

My team at Imperial College has provided invaluable counsel and guidance: Yvonne Pinto, Calum Handforth, and Sara Delaney, who were with us for various periods, and the core team, Jo Seed, who has provided much support, Katy Wilson, who did a great deal of the research for the book and wrote many sections, and my deputy, Liz Wilson, who is a continuing source of wisdom and practicality.

I am grateful to Heidi S. Lovette, science acquisitions editor at Cornell University Press, for her support and enthusiasm and also to the team at Cornell University Press for their ongoing dedication and assistance. I would also like to thank Ros Morley and Sarah Starkey for their work in the production of the book.

Editorial Note

For supplementary material, please visit www.canwefeedtheworld.org.

Regions

The *developed regions* (as per UN regional country groupings shown in Plate 1) comprise Europe (including the transition countries but not the Commonwealth of Independent States [CIS]), Australia, Canada, Japan, New Zealand, and the United States (US), shown in maroon.

The CIS (Commonwealth of Independent States), which include the territories of the former Soviet Union in Europe and Asia, is sometimes included in the developing regions.

However, in this book the *developing regions* are the rest of the world and can be subdivided into:

Middle-income countries—such as Brazil, Vietnam, and South Korea, and including the *emerging economies* or BRICs (a term derived from the first letters of Brazil, Russia, India, and China but now used generally to describe countries with rapidly emerging industry and improving per capita wealth);

Low-income countries—such as Kenya, Ghana, and Honduras, and including the 48 *least developed countries* such as Cambodia, Mali, and Haiti;

Fragile states include those who are conflict ridden or recently emerging from conflict such as Afghanistan, Sierra Leone, and Nepal.

Groups

G7 Finance ministers from France, Germany, United Kingdom, Italy, Canada, United States, and Japan (European Union [EU] also present)—established in 1975

G8 Heads of government for the G7, plus Russia established in 1997

G20 Finance ministers and central bank governors—Argentina, Australia, Brazil, Canada, China, European Union, France, Germany, India, Indonesia, Italy, Japan, Mexico, Russia, Saudi Arabia, South Africa, Republic of Korea, Turkey, United Kingdom, United States—established in 1999

Measures

Dollars ($) are U.S. dollars
1 billion = 1,000 million
1 trillion = 1,000 billion
1 kg = 1 kilogram = 2.2 pounds (lbs)
1 ton = 1 tonne = 1 metric ton
= 1,000 kg = 1 million grams (Mg)

1 ha = 1 hectare = 10,000 square meters
= 2.47 acres
1 Mha = 1 million hectares (Mha)
1 km = 1 kilometer = 0.62 miles
calories = kilocalories (kcal)
ppb = parts per billion

Grain means cereals, e.g., wheat, barley, rice, maize, oats, sorghums, millets, and other coarse grains. (It does not include grain legumes or pulses.)

minimum dietary energy requirement (MDER or r_L)

Institutional Acronyms

CCAFS CGIAR Research Program on Climate Change, Agriculture and Food Security
CGIAR Consultative Group on International Agricultural Research (Washington, DC)
CIAT Centro Internacional de Agricultura Tropical (Cali, Columbia)
CIFOR Center for International Forestry Research (Bogor, Indonesia)
CIMMYT Centro Internacional de Mejoramiento de Maiz y Trigo (Mexico City, Mexico)
CIP Centro Internacional de la Papa (Lima, Peru)
DFID Department for International Development (London)
FAO Food and Agriculture Organization of the United Nations (Rome)
IARCs International Agricultural Research Centers (funded by the CGIAR)
ICARDA International Center for Agricultural Research in the Dry Areas (Aleppo, Syria)
ICLARM World Fish Center (formerly International Center for Living Aquatic Resource Management [Penang, Malaysia])
ICRAF World Agroforestry Centre (formerly International Center for Research in Agroforestry [Nairobi, Kenya])
ICRISAT International Crops Research Institute for the Semi-Arid Tropics (Patancheru, Andhra Pradesh, India)
IDS Institute of Development Studies (Brighton, UK)
IFAD International Fund for Agricultural Development (Rome)
IFPRI International Food Policy Research Institute (Washington, DC)
IIED International Institute for Environment and Development (London)
IITA International Institute of Tropical Agriculture (Ibadan, Nigeria)
ILRI International Livestock Research Institute (Nairobi, Kenya)
IMF International Monetary Fund (Washington, DC)
IPC International Planning Committee for Food Sovereignty
IPCC Intergovernmental Panel on Climate Change
IRRI International Rice Research Institute (Manila, Philippines)
ISAR Institut des Sciences Agronomiques du Rwanda
IWMI International Water Management Institute (Colombo, Sri Lanka)
KARI Kenya Agricultural Research Institute
NGOs Nongovernmental organizations
MYRADA Mysore Relief and Development Agency (Bangalore, India)
OECD Organisation for Economic Co-operation and Development (Paris)
UN United Nations (New York)
UNEP United Nations Environment Programme (Nairobi, Kenya)
UNHCR United Nations High Commissioner for Refugees (Geneva)
UNICEF United Nations Children's Fund (New York)
UNRISD United Nations Research Institute for Social Development (Geneva)

USAID U.S. Agency for International Development (Washington, DC)
USDA U.S. Department of Agriculture (Washington, DC)
WFP World Food Programme (Rome)
WHO World Health Organization (Geneva)
WTO World Trade Organization (Geneva)

One Billion Hungry

Part I

1

Acute and Chronic Crises

> On Wednesday, police cleared away torched cars and other debris left by two days of looting and rioting. But helicopters circled the air amid black smoke rising from intersections as protesters continued to set tires ablaze, and gunfire was heard throughout Pétionville . . .
>
> Associated Press, Haiti, April 9, 2008[1]

Hunger (from the Old English *hungor*) is an evocative, old Germanic word meaning "unease or pain caused by lack of food, craving appetite, debility from lack of food."[2] In the developed countries it is a feeling of slight discomfort when a meal is late or missed. By contrast, in the developing countries hunger is a chronic problem. Television images convey the realities of hunger—emaciated and starving children—in war-torn countries or in the aftermath of droughts, floods, or other calamities. Yet for a billion people—men, women, and children—hunger in the developing countries is a day-to-day occurrence, both persistent and widespread.

Food security, which implies the absence of hunger, is one of those apparently straightforward concepts that appears amenable to common-sense definition. But, somewhat surprisingly, it has been the subject of much debate. A review by staff of the Institute of Development Studies (IDS) at Sussex University identified some two hundred different definitions.[3] It is a highly politicized concept (I discuss this further in Chapter 4).[4] Nevertheless, in 1986 after further debate, the World Bank adopted the following definition which, with various elaborations, is now largely accepted: "Food security is access by all people at all times to enough food for an active, healthy life."[5]

The controversy over the definition arises, in part, because food security operates at many different levels and over different time scales: It can apply to the globe as a whole, to a region such as Sub-Saharan Africa, or to an individual country, community, or household. These different levels relate only very loosely to each other. A country can be food secure, but a household may not. Sometimes lack of sufficient food can be temporary, although devastating; at others, it is persistent and seemingly intractable.

Mrs. Namarunda

How chronic food insecurity comes about and what it means in practice is well illustrated by the conditions under which an African woman farmer, such as Mrs. Namarunda (who represents a composite of situations existing in Africa), struggles to feed herself and her family (Box 1.1).

Although Mrs. Namarunda lives at the end of a poorly maintained track, far from a town or markets, she is not immune to the events in the larger world. The high price of food affects her; the small sums her son irregularly sends her are insufficient to keep her family adequately fed.

Box 1.1 A One-Hectare Farm in Kenya, Africa

Several years ago, Mrs. Namarunda's husband died from HIV/AIDS. Her eldest son inherited the family farm, a single hectare running up one side of a hill, in the Siaya district near Lake Victoria. The soils are moderately deep and well drained, but they are acidic, highly weathered, and leached. Mrs. Namurunda's first son married and moved to Nairobi, where he is a casual laborer with children of his own.

Mrs. Namarunda was left on the farm—still officially owned by her absent son—with four younger children and the responsibility to produce food, fetch water, gather fuel, educate the children, and care for the family. But shortages of almost everything—land, money, labor, plant nutrients in soil exhausted from many years of continual crop production—mean that she is often unable to provide her family with adequate food. The two youngest children suffer from undernourishment and persistent illnesses.

Like many others in Africa, Mrs. Namarunda's smallholding provides an "insecure" livelihood (Figure 1.1). Fertilizer is expensive and she can't get credit, so she starts each growing season with a maximum potential harvest of only about 2 tons from mixed cropping on her 1 hectare of land. To survive, her family requires a harvest of about 1 ton, so if everything goes right and the maximum harvest is achieved, it would be sufficient to meet their needs and to generate a modest income. But, during the course of every growing season, she faces innumerable threats to her crops that reduce her yields.

Weeds are her most persistent and pervasive problem. It takes 40 to 50 days of weeding each crop, by her and the children, to keep the weeds under control. Her staple crop, maize, is attacked by:

- *streak virus*, where leaves develop long, white, chlorophyll-depleted lesions
- *Striga*, the parasitic weed that sucks nutrients from the roots and poisons its host
- *boring insects*, which weaken the stem
- *fungi*, which rot the ears that do develop, before and after harvest

(*continued*)

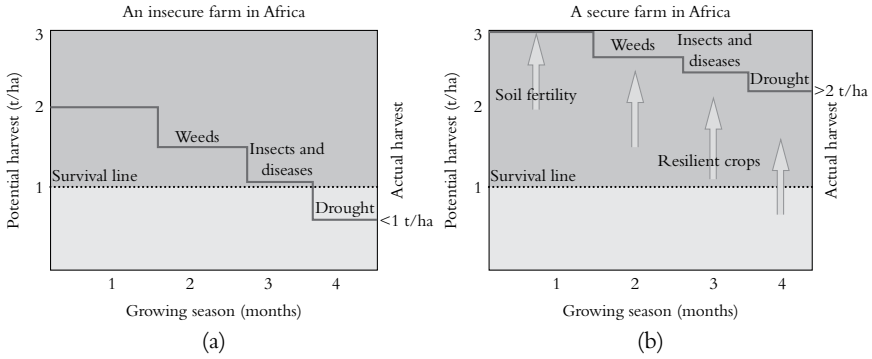


Figure 1.1 Harvest from an insecure and a secure farm.[6]

Mrs. Namarunda has tried growing cassava as an "insurance crop." But it, too, was attacked, first by mealy bugs and green mites, and then it was devastated by a new, supervirulent strain of African cassava mosaic virus.

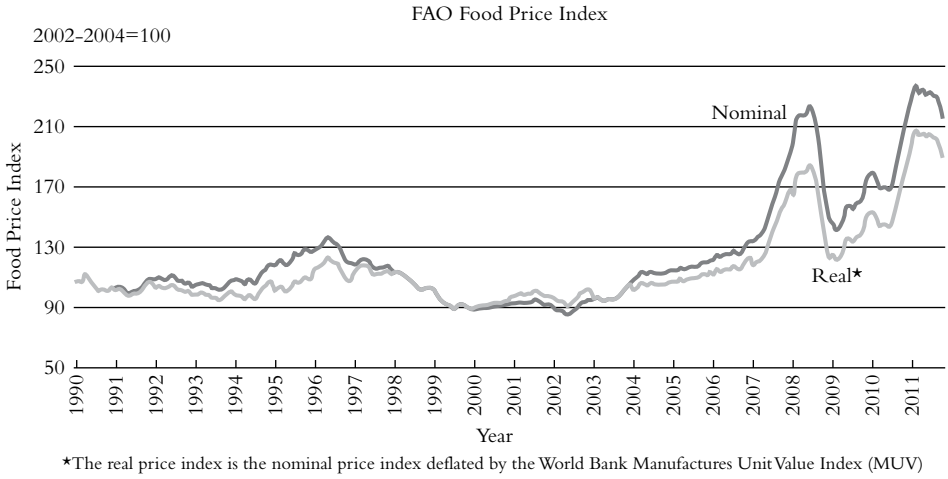
The banana suckers she obtained from neighbors were already infected with weevils, nematodes, and the fungal disease Black Sigatoka when she bought them. Her beans, which are intended as a source of protein for the family and nitrogen for the soil, suffer from fungal diseases that rot the roots, deform leaves, shrivel pods, and lower nitrogen fixation. She also faces periodic drought that reduces yields. At the end of each season, what she actually harvests is usually less than 1 ton. She and her children are often hungry, and there is no money for schooling or health care.

Food Price Spikes

When food prices rose in 2007 there were street demonstrations in Mexico City, protesting at the high price of tortillas. Subsequent protests, many associated with violence, erupted throughout Latin America, Africa, and Asia. The prime minister of Haiti was deposed following the riots in 2008. Even in Italy there were popular demonstrations triggered by the rise in the price of pasta.

The Food and Agriculture Organization of the United Nations (FAO; based in Rome) keeps track of the prices of basic food staples. They began to rise in late 2006, apparently following a rise in oil prices.[7] Initially it looked like a new, maybe long-term, trend but eventually in mid-2008 the food prices peaked and fell down again (Figure 1.2). To some extent it was a classic, short-lived commodity price spike:

first, a commodity becomes, or is perceived to be, scarce;
second, prices rise;



*The real price index is the nominal price index deflated by the World Bank Manufactures Unit Value Index (MUV)

Figure 1.2 The FAO Food Price Index as from 1990 to 2011.[8]

third, producers respond by producing more of the commodity and finally, prices fall.

But while food prices fell at the end of 2008, they were still 20 percent higher than 2006 prices and were growing again in 2009, tracking the 2007 rise. They leveled off in 2010 and then rose again, producing another spike. People in the developed countries felt the consequences, with higher food prices compounding the effects of the economic recession, but for the developing countries the outcomes were devastating. Grain prices remained high, even after the peak.[9] The FAO and the Organisation for Economic Co-operation and Development (OECD; based in Paris) estimated that, globally, average crop prices would be 10 to 20 percent higher in real terms relative to 1997–2006 for the next ten years, while for vegetable oils they would be more than 30 percent higher.[10]

I discuss how such food price volatility can be dampened in Chapter 4. Yet the spikes were not simple transitory events. They grew out of an underlying chronic crisis of global food insecurity and made this insecurity deeper and probably more persistent. According to the FAO, by the end of 2009 the number of chronically hungry people had risen to 1.02 billion, greater than at any time in human history.[18]

Box 1.2 The Causes of the Food Price Spikes

There has been much analysis of the proximal causes of the 2007–2008 food price spike.[11] A major factor was the fall in grain stocks. Demand had been steadily rising, and, while supply was also growing, adverse weather produced a run of poor harvests, enough to take stocks to dangerously low levels. Global consumption of grains and oilseeds exceeded production in seven of the eight years since 2000. The Australian drought and other weather events led to the very poor harvest of 2006. By 2007, stocks were only 14 percent of use (Figure 1.3). Importing countries became anxious about satisfying their future food needs and the prices began to spike.

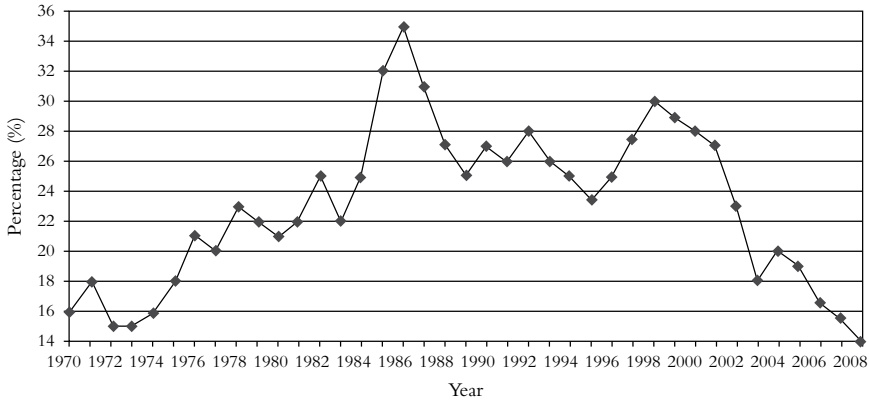


Figure 1.3 The stock to use percentage for all grains and oil seeds.[12]

The situation was compounded by speculation and by export bans imposed by Argentina, India, Kazakhstan, Pakistan, Ukraine, Russia, and Vietnam.[13] For rice, where the market is very thin (only 7 percent or 30 million tons of total production is traded), there was resort to panic buying. The Philippines purchased large amounts for domestic consumption at very high prices. Another factor was the depreciation of the U.S. dollar; some countries had accumulated considerable dollar reserves and were able to purchase large stocks of grain.[14] The energy price rise also played a role, as did the shift of land to biofuel production, especially in the United States.[15]

The subsequent spike in 2010 was triggered by a severe heat wave in Russia, which forecast a 30 percent reduction in the wheat harvest compared with the previous year, leading to an export ban that extended into 2011. Writers in such journals as the *Economist* and the *Financial Times* said there was plenty of wheat in stock and little to worry about[16]—"prices are unlikely to surge to the all time highs of 2007-08."[17] But then Egypt and other North African states purchased large quantities of wheat and prices began to rise. Various newspaper stories, ranging from possible locust outbreaks

(*continued*)

Box 1.2 (*continued*)

in Australia to heavy rains in Canada, heightened perceptions of impending shortage and future price rises. The actual and perceived shortages were enough to trigger a spike that eventually peaked above the 2007–2008 spike. Unlike that spike, the rises were not confined to food staples but extended initially to cotton and then to palm oil, cocoa, sugar, and rubber.

The Chronic Crisis

This book is concerned with three challenges: (1) the probability of repeated food price spikes and a continuing upward trend in food prices; (2) the persistence of a billion or more people suffering from chronic hunger; and (3) how we will feed a growing global population in the face of a wide range of adverse factors, including climate change. How will these challenges be manifest, both globally and in the developing countries, in the lives of the poor? What are their drivers? How do they operate at global, regional, and local levels? What can be done to mitigate or reverse their effects? Where do the barriers to action and the opportunities lie? How can we utilize the social and natural sciences to find solutions? How can we harness both government and the private sector to ensure solutions are applied on a large enough scale to achieve significant impact?

In the rest of this chapter I briefly look, in turn, at the fundamental drivers of

- the *increasing demand for food:* population increases, rising per capita incomes, and the competing demand for biofuel crops; and
- the *deficiencies in supply* caused by rising input prices, land and water scarcity and deterioration, slowing productivity gains and climate change.

Rising Populations

The first key driver of demand is the growth in the world's population.[19] According to the latest United Nations (UN) estimates, the global population is set to rise from close to 7 billion in 2010 to about 9.15 billion by 2050.[20] Then it is expected to stabilize. Virtually all of the additional 2.15 billion will be in the developing countries. The good news is that over forty percent of the world's population is living in countries where the fertility rate is at or under 2.1. The *fertility rate* represents the number of children an average woman is likely to have during her childbearing years, which in conventional international statistical usage is ages 15 to 49. A rate of 2.1 represents the *replacement level*; that is, the level of fertility at which a couple has

only enough children to replace themselves, taking account of daughters who die before childbearing years.[21] For countries with this fertility rate, population growth will slow down and eventually stabilize.

The UN expects fertility in the less-developed regions as a whole to drop from 2.73 children per woman in 2009 to 2.05 in about 2050, and the reduction projected for the group of forty-eight least developed countries is even steeper: from 4.39 children per woman to 2.41 children per woman.[22] This optimism is based on the experience of recent decades. Today's fertility decline in developing countries is similar to the European experience during nineteenth and early twentieth century industrialization, but even faster.[23] In Bangladesh, for example, the rate halved from 6.0 in 1980 to 3.0 by 2000—a mere 20 years.

There is a close link between standard of living and fertility rate. Fertility starts to drop at an annual income per person of $1,000 to $2,000 and falls until it hits the replacement level of 2.1 at an income per head of $4,000 to $10,000 a year. Thereafter fertility continues at or below replacement level. The link between living standards and fertility also exists within countries. India's poorest state, Bihar, has a fertility rate of about 4.0, while the richer states of Tamil Nadu and Kerala have rates below 2.0. In effect there are two interlinked transitions: one demographic, from high growth to population stability, and the other developmental, from an agrarian to an industrial society.[24]

Nevertheless, the fertility rate decline has not been universal. In many Sub-Saharan countries fertility rate declines have stalled at rates over 5.0 after gradually decreasing for several years.[25] The reasons are complex, but a common feature appears to be the decreased funding for family planning programs. According to data from thirty-one countries, on average 30 percent of women in Sub-Saharan Africa have an unmet need for modern family planning methods, a proportion that has not declined in the last decade.[26] In nineteen of these countries, it is as high as nearly 50 percent. If fertility were to remain constant at current levels, the population of less-developed regions would increase to 9.8 billion in 2050 instead of the projected 7.9 billion.[27]

A popular misconception is that providing the developing countries with more food will serve to increase populations; in other words, it is a self-defeating policy.[28] The more food women have, the more children they will have and the greater will be their children's survival, leading to population growth, so goes the argument. However, the experience of the demographic transition described above suggests the opposite. As people become more prosperous, which includes being better fed and having lower child mortality, the fewer children women want. Providing they then have access to family planning methods, the fertility rates will drop and the population will cease to grow.[29]

Rising Per Capita Incomes

A population rise to 9.15 billion—that is, an increase of about 30 percent—will require increased food production of a similar amount.[30] However, this assumes no change in consumption patterns or in the proportion of the population who remain malnourished and outside the market. It also ignores the many threats to food security we are facing.

Consumption patterns are changing fast. In recent decades, per capita incomes have increased in most countries of the world, leading to a greater demand for improved quality and, to some extent, larger quantity of food.

The growth in per capita incomes in the high-income OECD countries has been over threefold in constant U.S. dollars since the 1960s. Income growth has been even faster in East Asia and the Pacific (over sixfold), although from a much lower base. It has doubled in the Middle East and North Africa. Only in Sub-Saharan Africa has there been little growth (Figure 1.4).[31]

With rising incomes, growing urbanization, and exposure to so-called diet globalization, a significant proportion of people in both the developed and developing countries are buying more processed and higher value foods, while reducing their purchases of raw agricultural commodities. In Asia, the consumption of traditionally "Western" foods—wheat and wheat-based products, temperate-zone vegetables, and dairy products—is growing, while per capita consumption of rice is declining.[33] Per capita rice consumption in India, Indonesia, and Bangladesh peaked in the early

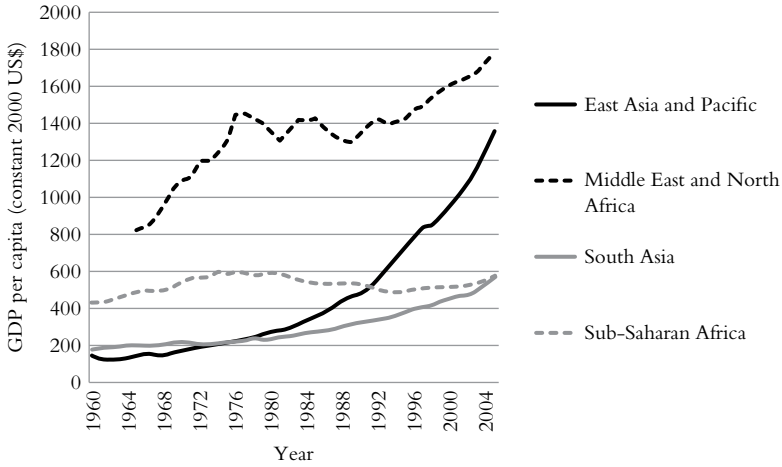


Figure 1.4 Growth in per capita incomes in low- and middle-income countries.[32]

1990s and has since fallen. Global rice consumption is also projected to fall from 441 million tons in 2010 to 360 million tons in 2050.[34]

One of the most significant outcomes has been the so-called Livestock Revolution, which many argue is having as important an effect on developing countries as the Green Revolution (see Chapter 10).[35] In recent decades, developing countries' meat consumption has been growing at over 5 percent per annum, and that of milk and dairy products at 3.5 percent to 4 percent.[36] The most dramatic increase has been in China, where meat consumption has risen to about 60 kg per person per year in 2005 from 20 kg per person per year in 1985.[37]

The correlation between income and consumption of meat and dairy products is strong, although religious and other cultural practices produce significant diversity. In Figure 1.5, China is above the line, showing the relationship between income and meat consumption, primarily because of the traditional consumption of pig meat (half of all the pig meat consumed in the world), and India is beneath, reflecting the continuing high levels of religious-based vegetarianism.[38]

Livestock, especially those reared intensively in confined forms of housing, are very demanding of cereals, oilseeds, and other feedstuffs. As a rule of thumb, it takes about eight kg of grain to produce one kg of meat product, although there is a considerable range from production on grazed grassland where little, if any, of the feed is brought in, to raising the livestock in stalls where the feed is entirely cereals and oilseeds.

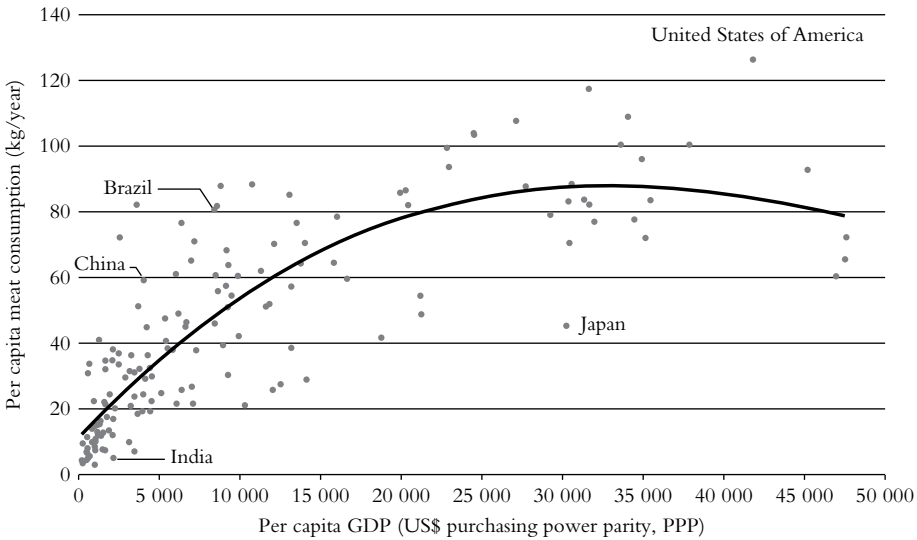


Figure 1.5 The relationship between meat consumption and per capita income.[39]

It is reasonable to expect that the Livestock Revolution will generate a greatly increased demand for feed, although this has not been borne out by recent history.[40] In the future that may change. China purchased 35 million tons of soybeans for its livestock and poultry industry in 2007, increasing oilseed prices. Although a one-off purchase, it may be a harbinger of the longer run impact of future demand.

The FAO expects there to be a slowdown in growth of meat consumption after 2030 but believes production will need to double from about 230 million tons in 1999–2001 to over 460 million tons by 2050, the great bulk of which will be produced in the developing countries.[41] As this happens, developing countries will shift from free-range to more intensive stall-fed systems of production, thus greatly increasing the demand for feed.[42] An estimate for 1999–2001 put the total feed use of cereals (note this excludes oilseeds) at 666 million tons, or 35 percent of world total cereal use. By 2050 this is likely to have doubled.[43]

Demand for Biofuels

Biofuel crops, or crops grown for products that can be converted to ethanol, biodiesel, or other fuels, can, in the right circumstances, reduce carbon and other greenhouse gas emissions (See Chapter 16). However, for many countries, especially the United States and China, the prime objective may be to reduce dependency on imported oil and petroleum. These objectives have driven a rapid rise in biofuel crop production in the developed countries and in some developing countries, notably China and Brazil. Global ethanol production has increased from under 20 billion liters in 2000 to around 85 billion liters in 2010. Global biodiesel production increased tenfold to almost 11 billion liters between 2000 and 2008, and in 2009 was almost 17 billion liters.[44]

The impact of this surge on the 2007–2008 food price spike is the subject of debate.[45] Donald Mitchell, the senior commodity economist at the World Bank, put the contribution as high as 75 percent (i.e., the greatest single cause of the spike) because growth in the demand for maize both for consumption and biofuels was not matched by growth in production (resulting in a shortfall of 27 million tons) and, as a consequence, a doubling of maize prices. However, Mark Rosegrant of the International Food Policy Research Institute (IFPRI) concluded the effect was much less, at under 40 percent.[46]

The discrepancies arise, according to Peter Timmer of Stanford University, because the estimates do not fully capture the linkages between maize, which is the primary grain used to make ethanol, and other commodities such as soybeans, wheat, and other feed grains. Thus, the 7 percent increased planting of maize in 2007 led

to reductions in soybean acreage of 4 percent, as maize and soybeans tend to compete for land area, increasing the price of soy oil.[47] China is not a significant importer of maize, but it is a huge importer of vegetable oils, and the higher price of soy oil in 2007 caused a switch to palm oil, which, in turn, increased in price.[48] The increased growing of biofuel crops is likely to make land substitution between maize, soybean, and wheat more frequent and intense. Prices of these crops will move together and also with dairy and meat products. The same phenomenon will apply to biodiesel crops such as oilseed rape.

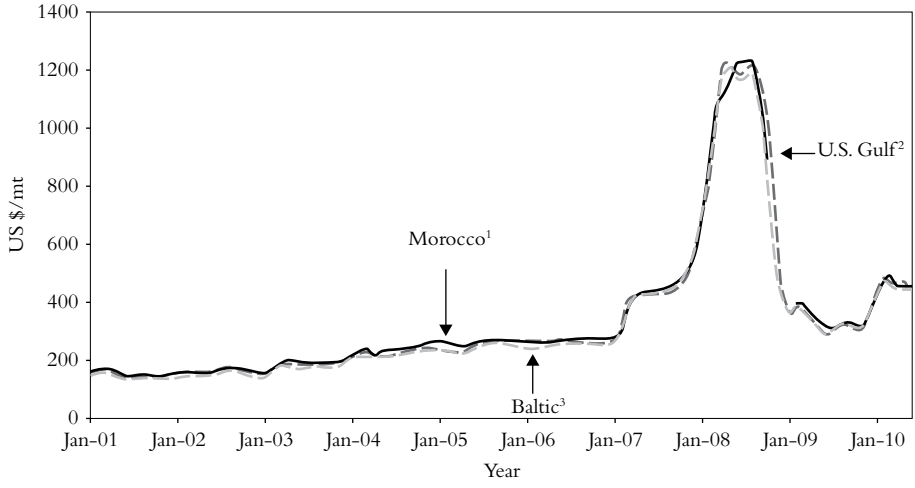
The range of estimated impacts of biofuels on global food production and prices has also been linked to different model assumptions regarding, for example, land availability, the substitutability of biofuels and food, and associations between food and energy markets.[49] Biofuels and the poor, a research project of IFPRI funded by the Bill & Melinda Gates Foundation, estimated that biofuel development in the United States, European Union, and Brazil will cause an increase in the world average export price of maize, soybean, other oilseeds and sugar of 17.7 percent, 13.6 percent, 27.6 percent, and 11.3 percent, respectively, by 2020.[50] The bottom line is that growing crops for biofuels will tend to increase the prices of basic food crops; biofuel production needs to be decoupled from food production either through policies or through cellulosic technologies.

I will discuss biofuel crops more fully in Chapter 16. For the rest of this chapter, I want to turn to the key long-term factors affecting supply.

Rising Oil and Fertilizer Prices

Rising oil prices were one of the key elements of the 2007–2008 food price spike. Indeed, their rise was a precursor to the food price increase. Their effect was, and continues to be, felt through the demand for biofuels, through the increased costs of transportation that affects both agricultural input and output prices, and in the manufacturing costs of fertilizers and other synthetic chemical inputs. In the developed countries the combined energy, chemical, and fertilizer costs typically account for a high proportion of crop production costs (16 percent for soybean, 27 percent for wheat, and 34 percent for maize in the United States).[51] Since the United States accounts for 40 percent of global grain exports and 25 percent of the global oilseeds exports, these costs are quickly reflected in global food prices.[52]

Energy-based inputs are also high in the irrigated lands of developing countries and growing rapidly where water is being depleted and water tables declining.[53] There, and on rain-fed lands, use of fertilizers and pesticides is often excessive and inefficient. Nevertheless, fertilizer and other inputs are crucial for small farmers trying to achieve higher yields of food crops. At the time of the food price spike, one



1. Derived from *Green Markets*. 2. Derived from *Fertilizer Week*.
3. Derived from *FMB Weekly Fertilizer Report*.

Figure 1.6 Increasing fertilizer prices (Diammonium phosphate, freight on board, monthly prices are all represented by a single line).[54]

of the biggest fertilizer price increases was in diammonium phosphate (DAP), a commonly used source of nutrients in developing countries (Figure 1.6). Its price rose nearly sixfold in early 2008, partly due to the energy prices involved in the production of the ammonium, but also because of shortages in both sulfur and phosphate, key elements in its manufacture. Although the subsequent fall was significant, the price of DAP fertilizer began to grow again from the middle of 2009.

The Scarcity of Land

Before 1960 greater food supplies were obtained by taking more land into cultivation; since then raising crop yields has been far more important.[55] However, there has been a continuing controversy over how much potential cultivable land remains for use. In the 1990s three apparently thorough analyses were conducted using the data provided by the FAO's *Soil Map of the World* and their annual *Production Yearbooks* and subsequent *Statistical Yearbooks*.[56] These indicated considerable areas of land available for cultivation: Asia-Pacific with 778 million hectares (Mha) available (27.1 percent of total land area), Sub-Saharan Africa with 1.2 billion ha available (45.8 percent of total), and South and Central America with one billion ha (50.1 percent).[57]

Yet these seem to be overestimates. The results have been trenchantly criticized by Anthony Young, who has a long and extensive experience of soil and land surveys.[58] He believes the estimates suffer from the following flaws:

- Overestimation of cultivable land (not accounting for features such as hills and rock outcrops when the maps are reduced in scale)
- Underestimation of presently cultivated land (illegal land occupation; e.g., forest incursions, not recorded)
- Failure to take sufficient account of land required for purposes other than cultivation (underestimates of human settlements and industrial use)

A more recent FAO analysis in 2000 accepts these criticisms as possibly valid and acknowledges that much of the cultivable but uncultivated land is under rainforest or needed for purposes such as grazing land and ecosystem services.[59] Probably the most telling data is the area harvested over time. Total global cropland has increased by only 10 percent over the past fifty years, while population has grown by 110 percent.[60] Given the pressures to increase food production, we would expect to see much greater land expansion if it were readily available. The only exceptions are for oil crops (Figure 1.7). Soybeans and oil palms have each increased by over 300 percent in area and by over 700 percent and over 1,400 percent, respectively, in production over the past fifty years. Presumably this is a result of clearing the Cerrado in Brazil and rainforests in the Amazon, Africa, and Southeast Asia. Permanent meadows and pastures (the land used to grow herbaceous forage crops) have also increased somewhat, by nearly 9 percent in area from 1961 to 2008.[61]

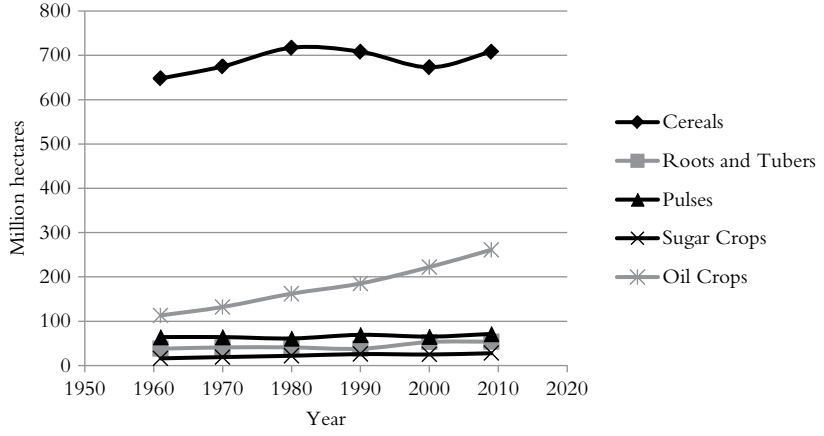


Figure 1.7 Trends in harvested area for selected food crops, million hectares.[62]

More land could be brought into cultivation by clearing the tropical rainforests, but this would be at the expense of biodiversity and would add considerably to greenhouse gas emissions (see Chapter 16). A further factor in the equation is the large amount of land being degraded as a result of erosion, loss of fertility, and desertification. According to an essentially qualitative *Global Assessment of Human Induced Soil Degradation (GLASOD),* about 300 million hectares (Mha), or 5 percent of the formerly usable land in developing countries, has been lost by severe soil degradation as of 1991; that is, more than has been brought into production.[63] The current rate of loss is claimed to be not less than five Mha per year.[64]

"Land Grabs"

Despite this apparent lack of available arable land, or maybe because of it, foreign or national investors have been purchasing large tracts of land in developing countries, often for the production of crops destined for export. In 2009, hedge funds and other speculators bought or leased almost 60 Mha of land in Africa.[65] Ethiopia, a country known for its famines and reliance on foreign aid, began to sell millions of hectares of land to investors in 2008.[66] The impacts of such investments on host countries are unclear given the diversity of land acquisition agreements and investor commitments. Increased investment may boost gross domestic product (GDP) growth and benefit economic development through employment and infrastructure. Indeed, the Ethiopian government claims land deals will bring in much-needed foreign currency and facilitate technology transfer from large agribusinesses to smallholder farmers.[67]

But while communities may benefit from these developments, access to land crucial for food security may be taken from local people, particularly because land users in developing countries are often marginalized in the land ownership system.[68] Although large areas of land "look empty," they are not. Often local communities have a range of traditional rights over the land, and many countries do not have legal or procedural mechanisms to protect these rights. Furthermore, a lack of transparency and regulation can foster corruption. Millions of local people have already been displaced from areas purchased by investors, the majority without any form of compensation.

Large-scale land grabs could also create environmental conflicts if incentives such as priority rights over water are offered to encourage investment or if inappropriate resource-intensive farming practices are employed over large areas. In a 2011 review of twelve land deals in Africa, such incentives and long-term rights to land were, in some cases, offered for very little in return (e.g., jobs or investment in infrastructure) and with little consideration of social and environmental issues.[69] Neverthe-

less, large-scale land acquisition in Africa will likely increase in the future, in part spurred by concerns over food security, competing land uses, rising food prices, and human expansion in investor countries.[70]

Scarcity of Water

Water is, of course, crucial to agricultural production and, like land, is similarly in short supply and for similar reasons—overuse, inefficient use, and degradation through pollution.[71] Today, much of Asia's grain harvest comes from the irrigated, annual double- and triple-crop, continuous rice systems in the tropical and subtropical lowlands of Asia and from irrigated, annual rice-wheat, double-crop systems in northern India, Pakistan, Nepal, and southern China. Large areas of this land and of the Middle East and North Africa are now maintaining irrigated food production through unsustainable extractions of water from rivers or the ground.[72] In China the groundwater overdraft rate exceeds 25 percent, and the rate is over 56 percent in parts of northwest India.[73] Groundwater levels in India have declined by nearly 18 cubic km per year over the past decade.[74] Much of this excessive water use results from water pumps using subsidized or free electricity.[75]

By contrast, Sub-Saharan Africa has large untapped water resources for agriculture. Only 4 percent of cultivated land (5.3 Mha) is irrigated, of which 70 percent is in Madagascar, South Africa, and Sudan. The potential exists to bring an additional 20 Mha of land under irrigation but, so far, technical, financial, and socioeconomic constraints have slowed this expansion. At the same time, almost a quarter of the African population lives in water-stressed countries, and the share is rising.

Many river basins in the world do not have enough water to meet all the demands; about a fifth of the world's people, more than 1.2 billion, live in areas of physical water scarcity. Rivers are drying up, groundwater levels are declining rapidly, freshwater fisheries are being damaged, and salinization and water pollution are increasing.

I discuss soil and water conservation in more detail in Chapters 13 and 14. In combination, declining amounts and quality of cultivable land and of utilizable water will make agriculture less productive and food more expensive.

Slowing Productivity Increases

As I stated earlier, global consumption of grains and oilseeds exceeded production in seven of the years between 2000 and 2008. This was partly due to declines in production resulting from extreme climatic events. Deteriorating land and water supplies may have had an effect. But there is also some evidence of declines in the rate of growth in yields of the major cereal and oilseeds. Figure 1.8, which measures the

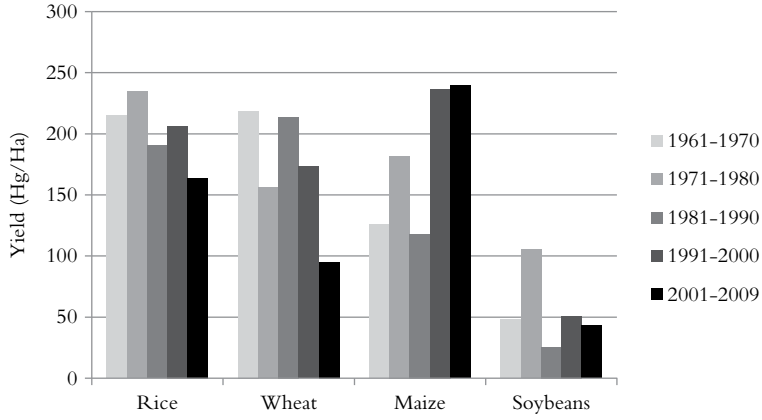


Figure 1.8 Average decadal increase in yield for rice, wheat, maize, and soybeans between 1961 and 2009.[76]

average decadal increase in yields, shows the high rates of growth in the Green Revolution years of the 1960s and 1970s for all four crops—rice, wheat, maize, and soybeans. Subsequently there have been declines in rates of growth for rice and wheat, although yields of maize and soybeans have both grown strongly in the past two decades.

The reasons for the decline are not clear, but one factor may have been the reduced investment in agricultural research and development by government.[77] Low stable food prices since the late 1970s have led to complacency about global food supply and to a reduction in research and development (R&D) funding. There has also been a partly ideologically based belief that agriculture is best served by the private sector. Indeed, private funding of agricultural research has grown but has tended to focus on the crops of developed countries where the returns have been greater. This may be the reason why yield growth in maize and soybeans has been strong in recent years.

The Impact of Climate Change

In the longer term climate change is likely to have a bigger effect on food supply than any other factor. Moreover, agriculture will probably be affected more than any other economic sector in the developing countries (this may also be true for the developed countries).

As yet, the effects are localized and mostly lie within the range of natural climatic variation. The 2007–2008 price spike was not a result of climate change, although

there is some evidence that the severity of the catastrophic 2006 drought in Australia, which had a major effect on world grain supplies, was a consequence of global warming.[78] Australia naturally has a highly variable climate, and the 400,000 square mile Murray-Darling basin, the country's prime food-growing region, is subject to periodic severe droughts, largely influenced by the El Niño phenomenon in the Pacific (see Chapter 15). The year 2006 had the lowest rainfall for one hundred years, but it also experienced the highest temperature. The subsequent higher evapotranspiration coupled with drought resulted in even lower yields than might otherwise have occurred. This trend is likely to continue to have an increasingly adverse effect on Australia's crop production.

A similar phenomenon may have occurred in Russia in the summer of 2010. There, temperatures reached 5°C above the norm for July, leading to a doubling in Moscow's death rate and the burning of 30 percent of the nation's grain crops. Notably this was combined with the worst floods in eighty years in Pakistan, which killed over 1,600 people and submerged a fifth of the country, including about 14 percent of Pakistan's cultivated land. Global warming is expected to increase the frequency and duration of extreme weather events but climate models are not yet sophisticated enough to determine whether the abnormal airflow in the region in 2010 was the result of increasing greenhouse gas emissions.[79]

In Chapter 15 I will show how future climate change will have an increasingly adverse effect, worldwide, on food security and discuss what can be done to adapt.

Outline of the Book

In some respects this is a second edition of my book *The Doubly Green Revolution: Food for All in the 21st Century,* first published in 1997.

As I began to write I realized how much had changed in the intervening years. Most of the concepts and arguments remain as relevant today as then, in some cases more so. I have retained some of the elements of the original structure—after all agriculture retains some constant truths. Some of the original examples are present, where they remain pertinent. But the book reflects what has happened globally in the past few years—the growing number of people who are chronically hungry, the devastating impact of the food price spikes and the challenges we face if we are to feed a world subject to climate change. Alongside these challenges there have been enormous strides in technical achievements—revolutionary improvements in crop and livestock breeding, successes in application of ecological principles to agricultural production and real examples of exciting farmer innovation. Out of the concept of a Doubly Green Revolution has grown an emphasis on Sustainable

Intensification, partly based on the recognition of the increasing shortage of land and water and of the threats posed by climate change. As a consequence there are new chapters and much new material. In essence it is a new book.

The first part of the book sets out the context more fully. In Chapter 2, I describe what is meant by *hunger*, how we measure it, and how it relates to poverty. Chapter 3 discusses the lessons from the Green Revolution of the 1960s and 1970s, one of the great technological success stories of the twentieth century. In Chapter 4 I discuss the political economy of food security and agriculture and how it is changing over time.

In the second part, I introduce the concept of a Doubly Green Revolution (Chapter 5) and then discuss its various components: the challenges of sustainable intensification (Chapter 6), the need for appropriate research and development, including technologies (Chapter 7), and the creation of markets (Chapter 8).

Part 3 examines specific areas in detail. Chapter 9 discusses the role of breeding to produce better crop varieties and livestock breeds, drawing on the full range of traditional, conventional, and biotechnology approaches. Chapter 10 is devoted to the topic of the Livestock Revolution, examining the role of domesticated animals as important components of agricultural livelihoods. Chapter 11 explores the contribution of farmers as innovators.

In Part 4 I discuss some of the environmental challenges facing the Doubly Green Revolution. Chapter 12 focuses on pest control, arguing for a combination of integrated pest management techniques with the use of biotechnology. Chapters 13 and 14 discuss soil and water conservation, emphasizing the importance of restoring good organic matter to the soil and efficiently using available water resources. In Chapters 15 and 16 I analyze the relationship between climate change and agriculture, where agriculture is presented as both victim and culprit.

Finally in chapter 17 I summarize the previous chapters and return to the question posed in the title of the book: Can we feed the world?

2

What Is Hunger?

To die of hunger is the bitterest of fates.

Homer, *The Odyssey*[1]

Homer, in his epic poem *The Odyssey,* recounts how Odysseus and his companions have resisted the lure of the Sirens, sailed safely between Scylla and Charybdis, and have come to the island of Thrinacia where the "Sun-god's cattle and plump sheep graze." Odysseus has been warned the animals are not to be harmed, but his companions succumb to the temptation. "To die of hunger," declares Eurylochus "is the bitterest of fates." They kill the cattle and feast. No sooner have they set sail again than Zeus sends a hurricane as a punishment. All perish except for Odysseus.

Today, there are a billion people who, like Odysseus's companions, live in a world where food is relatively plentiful yet it is not available to them. If we were to add up all of the world's production of food and then divide it equally among the world's population, each man, woman, and child would receive a daily average of over 2,800 calories[2]—enough for a healthy lifestyle. But of course, food is not divided in this way (nor is income) and it is unrealistic to expect it will happen in the near, or even distant, future.

Unlike Odysseus's companions, many of the hungry are women and children. In addition to their own needs, women require an adequate diet if they are to give birth to, and raise, healthy children. More than 130 million children under five years of age are underweight; that is, they are well below the standard weight for their age. This represents nearly a quarter of the under fives in the developing countries.[3] Children are especially vulnerable from their time in the womb until the age of five. During this period of rapid physical and cognitive development, even short-term dietary deprivation can have lasting effects such as stunted growth, poor brain development, low educational attainment, and decreased life expectancy. Girls grow into women who give birth to sickly children, so that ill health and poverty are passed from one generation to the next. In this way, hunger and poverty are inextricably

linked, one resulting in the other, so creating a trap from which escape is very difficult.

In this chapter I address these questions:

- What is hunger and how is it measured?
- What is the balance between food supply and access to food?
- How does hunger relate to poverty?

The Nature of Hunger

Hunger comes in many forms: [4]

- *Malnutrition* is a general term for a condition caused by improper diet or nutrition. It means "badly nourished," but is more than a measure of how much people eat or fail to eat. Malnutrition results from inadequate intake of protein, calories, and micronutrients and is characterized by frequent infections and disease. As a consequence the physical function of an individual is impaired to the point where he or she can no longer maintain natural bodily capacities such as growth, pregnancy, lactation, learning abilities, physical work, and resisting and recovering from disease.
- *Undernourishment* is more specifically used to describe the status of people whose food intake does not include enough calories (energy) to meet minimum physiological needs for an active life.
- *Starvation* is an extreme form of these conditions, characterized by a "state of exhaustion of the body caused by lack of food."[5] When starvation is accompanied by increasing mortality on an epidemic scale, we usually describe it as a *famine.*

These conditions are difficult to measure, so indicators have been devised that simply measure the height or weight of an adult or child and compare it with the same measurement in a reference population of well-nourished and healthy people. Three such measurements—stunting, wasting, and underweight—are commonly used. In practice they are relatively accurate reflections of hunger (Box 2.1).

The Food We Need

Determining the amount of food we need is complicated.

First, the sources of calories are very different. A *food calorie* or *kilocalorie* (kcal) is defined as the energy needed to increase the temperature of 1 kilogram (kg) of water

Box 2.1 Practical Measures of Hunger[6]

- *Stunting* or low height for age

  Caused by long-term insufficient nutrient intake and frequent infections. It generally occurs before age two, and the effects are largely irreversible. These include delayed motor development, impaired cognitive function, and poor school performance.

- *Wasting* or low weight for height

  Is a strong predictor of mortality among children under five. It is usually the result of acute significant food shortage and/or disease and reflects a recent and severe process that has led to substantial weight loss, usually associated with starvation and/or disease. Wasting is often used to assess the severity of emergencies because it is strongly related to mortality.

- *Underweight* or weight-for-age of a child

  It is estimated that the deaths of 3.7 million children aged less than five are associated with the underweight status of the children themselves or of their mothers.[7]

Most famine reported in the media is the consequence of disasters such as drought, floods, war, or earthquakes, but disaster-related famine accounts only for 8 percent of hunger-related deaths; 92 percent result from chronic and recurring hunger. Child wasting tends to be a consequence of disasters because it measures intermittent, short-term, acute undernourishment. Chronic hunger is best captured by measures of child stunting.[8]

by 1°C and is used as a common measure of the energy that cells acquire from food through digestive breakdown and oxidation. There are several principal sources of food energy.[9]

- *Carbohydrates* are the main source of energy and hence play a key role in basic metabolism.[10] Each gram of carbohydrate provides about two kcals, and the World Health Organization (WHO) guideline is that carbohydrates should provide 55–75 percent of total energy.
- *Fats*, comprising saturated fats, monounsaturated fats, and polyunsaturated fats, are the second major dietary energy source. Some fats are also essential for proper growth and development in early life. Each gram provides about nine kcals. Dietary guidelines, backed up by scientific studies, generally advise that a diet high in saturated fats will lead to increased blood cholesterol levels and a higher risk of heart disease.[11] Ideally fats should provide 15–30 percent of total energy.

- *Proteins*, comprised of amino acids, are vital for both maintenance and growth.[12] The amount and quality of protein intake promotes good muscle and bone growth and overall optimal human health.[13] Some amino acids cannot be synthesized in the body and are, therefore, important constituents in the diet. Meat, fish, eggs, and dairy proteins show a greater digestibility than plant sources, such as pulses and nuts.[14] Ideally protein should provide 10–15 percent of total energy.

Second, individuals differ greatly in the calories they need.

Energy requirements vary with an individual's metabolism, physical activity, and growth needs as well as with the requirements for pregnancy and lactation. Important determinants are age, gender, body size, and composition. Adult men tend to require more calories than women. Children have much lower requirements, increasing with age up to a peak around adolescence. Adults over age sixty also have lower requirements. Within each range, the need is determined by body size and activity level (Table 2.1).[15]

From these needs it is possible to use a country's population composition broken down by age, gender, body size, activity level, and pregnancy status to work out its overall needs. The Food and Agriculture Organization of the United Nations (FAO) does this by determining for each country the minimum dietary energy requirement (MDER). This is the amount of energy needed for light activity and to maintain a minimum acceptable weight for attained height. The numbers of hungry are then those who get less than this amount.[17]

In some countries such as the Democratic Republic of Congo (DRC) and Eritrea the average food supply is less than the MDER, and the percentage suffering from undernourishment is very high (Figure 2.1). In other countries the average food supply is higher than the MDER, and the greater the difference, the lower the percentage undernourished. While many argue that hunger is not determined by how

Table 2.1 Energy needs in calories per day

Age	Men/boys	Women/girls
0–12 months	518–775 (beginning and end of range)	464–712
1–18 years	948–3410	865–2503
18–30 years	2100–4500 (depending on weight and activity)	1650–3850
30–60 years	2100–4200	1750–3400
Over 60 years	1700–3600	1550–3150

Source: FAO/WHO/UNU[16]

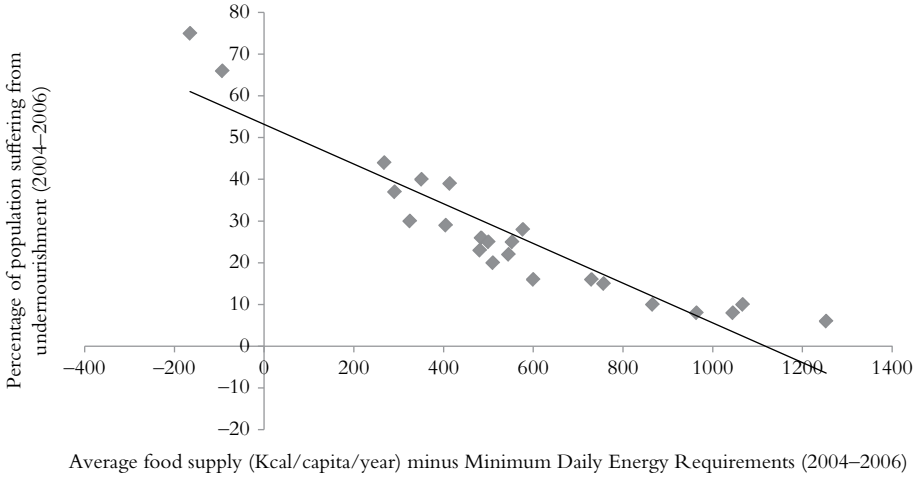


Figure 2.1 National proportion undernourished in relation to the gap between the Average Food Supply and the Minimum Daily Energy Requirement for 23 developing countries.[18]

much food is available but rather by how well it is distributed, these results indicate that the overall supply of food is a critical factor.

Not Just Calories

Poor nutrition is not just about a lack of calories. If a cereal diet supplies enough calories, it usually provides enough protein. Since, at least for the poor, cereals are the most important source of calories, there is a tendency for adequate nutrition to be equated with sufficient cereal production and consumption. Nevertheless, vegetables, fruit, livestock, and fishery products are also important sources of proteins, as well as vitamins and minerals. It is total food supply, measured in terms of quality as well as quantity, that determines whether people are adequately fed.

It is generally accepted that a well-balanced diet is essential, but for many years there has been a lively and, at times, acrimonious debate over where the balance lies and what components are most deficient. The focus of nutrition interventions has evolved from countering protein deficiency, to concern over protein-energy deficiency, and now to the prevention and treatment of micronutrient deficiencies.

In 1935, a Jamaican pediatrician, Dr. Cicely Williams, described a set of symptoms that she termed kwashiorkor that appeared to be related to lack of protein in the diet.[19] Children suffered from accumulations of liquid under the skin (edema), retardation in growth, wasted muscles, and some psychomotor damage.[20] The condition was especially prevalent in Africa (the name is of Ghanaian origin), in part because of the low protein content of the staple starchy foods.

Although there were no reliable data regarding the prevalence of kwashiorkor, the subsequent conventional wisdom was that protein deficiency was the main nutrition problem in the world, a view reinforced by a UN Protein Advisory Group that promoted the use of protein-rich food mixtures and by the shipment of vast quantities of dried skimmed milk to developing countries.[21] The terms *kwashiorkor* and *malnutrition* were often used interchangeably. Then, in 1963, Derrick Jelliffe, professor of Paediatrics and Child Health at Makerere Medical School in Uganda, pointed out that there were many other childhood conditions in addition to kwashiorkor. The main problem in all of these, he concluded, was a combination of calorie and protein deficiency.[22] The controversy intensified in the 1970s with the publication of an article on "the great protein fiasco" by Professor Donald McLaren, a nutritionist working at the American University of Beirut, in which he claimed the role of protein deficiency as a cause of child malnutrition had been overstated.[23] Others agreed that protein intakes seemed to be adequate in most countries, and the emphasis subsequently shifted to improving the health of women and children through growth monitoring and interventions such as oral rehydration therapy, breastfeeding, and immunization.[24]

The First 1,000 Days

In a five-part series on nutrition issued by the *Lancet* in 2008, adequate nutrition was found to be critically important during the first 1,000 days of a child's life (i.e., from conception to two years of age). Undernutrition during this crucial window increases the risk of early mortality and has largely irreversible long-term effects on health and on cognitive and physical development (Box 2.2).[25]

Micronutrient Deficiencies

The latest phase of this debate, partly reflected in the Scaling Up Nutrition (SUN) initiative (see Box 2.2), has been a focus on micronutrient deficiencies. Although the consequences of severe deficiencies—especially of vitamin A, iron, zinc, and iodine—have long been known, there has been little understanding that milder forms could be harmful and are widespread.[29]

Box 2.2 Interventions in the First 1,000 Days

Approximately 195 million children in the world are stunted. This is a third of all children who are under 5 years old. Of these, 90 percent live in just thirty-six countries, twenty-one of which are in Sub-Saharan Africa. In some African countries the proportion of children stunted is as high as 50 percent.

Undernutrition causes an estimated 3.5 million maternal and child deaths annually. As U.S. Secretary of State Hillary Clinton said at CARE's 2010 National Conference and Celebration, "These deaths are intolerable because they are preventable."

Direct nutrition interventions targeted at the 1,000-day window are cost-effective: the Copenhagen Consensus Panel, a panel of leading economists, found vitamin A and zinc supplementation to yield benefits worth $1 billion per year for an annual investment of $60 million.[26] In 2009, a World Bank study identified a package of thirteen interventions for the first 1,000 days, including promoting good feeding practices, providing micronutrients through supplementation or fortification, and therapeutic feeding. In return for implementing these interventions in the thirty-six highest burden countries (estimated to cost $11.8 billion annually) the lives of one million children annually would be saved. Despite this evidence, expenditure by major aid donors on nutrition totaled just $439 million between 2002 and 2007 (less than 1 percent of total bilateral development assistance).[27]

The Scaling Up Nutrition (SUN) framework, launched in April 2010, aims to set out a plan of action and to mobilize support for tackling child undernutrition. Several developing countries—including Bangladesh, Ghana, Malawi, Nepal, and Uganda—have indicated they wish to engage with the SUN agenda. The goal is that at least eight countries will start to receive intensive support for SUN by 2012.[28]

- *Vitamin A.* About 130 million preschool children suffer from vitamin A deficiency. As has been well established, lack of this vitamin can cause eye damage. Half a million children become partially or totally blind each year, and many subsequently die. As more recent research also has shown, lack of vitamin A has an even more serious and pervasive effect, apparently reducing the ability of a child's immune system to cope with infection. Trials in several developing countries where children under five were given vitamin A supplementation resulted in a 25 percent reduction in mortality from measles and diarrhea.[30]
- *Iron.* Lack of minerals in the diet can have equally severe effects. Iron deficiency is common in the developing world, affecting as many as a billion people. Iron deficiency is the leading cause of anemia (a reduction in the number of red blood cells below normal levels), which afflicts over 30 percent of women of childbearing age (15–49 years old) and 42 percent of all pregnant women. In Africa the figures are 47 percent and 57 percent, respectively. Iron-deficient